

CY 563: Transition Metal Chemistry of N-Heterocyclic Carbenes

Course Code	CY 563
Course Name	Transition Metal Chemistry of N-Heterocyclic Carbenes
L-T-P-C	3-0-0-3
Prerequisites	CY 531, CY 533 (only for BS students)
Intended for	UG/PG/PhD
Distribution	Elective
Mutual Exclusion	None
Approval	64th BoA (May 2026)

Preamble

This course, “Transition Metal Chemistry of N-Heterocyclic Carbenes,” is designed to provide BS-MS, M.Sc. and Ph.D. students the basic concepts and current topics in transition metal complexes featuring N-heterocyclic carbene ligands and their various applications in catalysis. This course emphasizes the huge impact of the N-heterocyclic carbene ligands in the coordination chemistry and in the field of homogeneous catalysis. A detailed insight including historic developments of stable carbenes, electronic properties, steric properties, coordination chemistry, small molecule activation and metal-complexes in catalysis will give a broad and concept-driven pedagogy to the students.

Course Modules

- **Module 1. Introduction to N-Heterocyclic Carbenes (NHCs): Synthesis and Electronic Parameters:** Introduction, Electronic structure and stabilization of NHCs, Historical development of stable carbenes, σ -Donor and π -acceptor properties, Steric parameters (buried volume, steric maps), Tolman Electronic Parameter (TEP), Methods for measuring donor properties, Synthesis of NHC Precursors, Preparation of Free NHCs, NHCs derived from four-, five- and six-membered heterocycles. (8 Hours)
- **Module 2. Emerging Classes of Carbene Ligands:** Cyclic (alkyl)(amino)carbenes (CAACs), Bicyclic (alkyl)(amino)carbenes (BICAACs), Mesoionic carbenes (MICs), Abnormal NHCs (aNHCs), Remote NHCs. A brief comparison of electronic properties and reactivity of the above-mentioned carbene ligands, Activation of small molecules. Formation of B-B and Al-Al multiple bonds with carbenes. (6 Hours)
- **Module 3. Group 1 and 2 and Early Transition Metal Carbene Complexes:** Carbene complexes of Group 1 and Group 2 (Li, Na, Be, Mg, Ca), Carbene complexes of Group 3-7 metals. Structure and bonding, Oxidation states and electronic effects, Metal–ligand cooperativity, Reactivity, Small molecule activation, Applications in catalysis (10 Hours)
- **Module 4. Late Transition Metal Carbene Complexes:** Carbene complexes of Group 8-12 metals. Synthesis, structure, characterization. Oxidation states and electronic effects, Metal–ligand cooperativity. Applications in catalysis, e.g., C-C, C-X bond forming reactions, metathesis reactions, functionalization of C-H bonds, hydroformylations reactions, hydroamination of alkenes, cross-coupling reactions, boration and silylation reactions. (14 Hours)

- **Module 5: Biologically Active Metal-NHC Complexes:** Synthesis, structure and characterization. Antimicrobial and anticancer properties of silver, gold and platinum-NHC complexes. (4 hours)

Text Books

1. *N-Heterocyclic Carbenes*, Edited by Silvia Diez-Gonzalez, RSC Publishing, 2010. <https://doi.org/10.1039/9781849732161>
2. *N-Heterocyclic carbene complexes of transition metals*, Robert Carroll, VDM Verlag, 2009.

References

1. *N-Heterocyclic Carbenes in Synthesis*, Edited by Steven P. Nolan, Wiley-VCH, 2006. <https://onlinelibrary.wiley.com/doi/book/10.1002/9783527609451>
2. F. E. Hahn, M. C. Jahnke, F. E. Hahn, M. C. Jahnke, *Heterocyclic Carbenes: Synthesis and Coordination Chemistry* Angew. Chem. Int. Ed. 2008, 47, 3122–3172.
3. Á. Vivancos, C. Segarra, M. Albrecht, *Mesoionic and Related Less Heteroatom-Stabilized N-Heterocyclic Carbene Complexes: Synthesis, Catalysis, and Other Applications* Chem. Rev. 2018, 118, 9493–9586.
4. E. Peris, *Smart N-Heterocyclic Carbene Ligands in Catalysis* Chem. Rev. 2018, 118, 9988–10031.
5. M. Melaimi, R. Jazzar, M. Soleilhavoup, G. Bertrand, *Cyclic (Alkyl)(amino)carbenes (CAACs): Recent Developments* Angew. Chem. Int. Ed. 2017, 56, 10046–10068.
6. D. J. Nelson, S. P. Nolan, *Quantifying and understanding the electronic properties of N-heterocyclic carbenes* Chem. Soc. Rev. 2013, 42, 6723–6753.
7. R. D. J. Froese, C. Lombardi, M. Pompeo, R. P. Rucker, M. G. Organ, *Designing Pd-N-Heterocyclic Carbene Complexes for High Reactivity and Selectivity for Cross-Coupling Applications* Acc. Chem. Res. 2017, 50, 2244–2253.
8. M. N. Hopkinson, C. Richter, M. Schedler, F. Glorius, *An overview of N-heterocyclic carbenes* Nature 2014, 510, 485–496.
9. S. Bellemin-Laponnaz, S. Dagorne, *Group 1 and 2 and Early Transition Metal Complexes Bearing N-Heterocyclic Carbene Ligands: Coordination Chemistry, Reactivity, and Applications* Chem. Rev. 2014, 114, 8747–8774.
10. H. Braunschweig, R. D. Dewhurst, *Boron–Boron Multiple Bonding: From Charged to Neutral and Back Again* Organometallics 2014, 33, 6271–6277.
11. S. Roy, K. C. Mondal, H. W. Roesky, *Cyclic Alkyl(amino) Carbene Stabilized Complexes with Low Coordinate Metals of Enduring Nature* Acc. Chem. Res. 2016, 49, 357–369.
12. V. Nesterov, D. Reiter, P. Bag, P. Frisch, R. Holzner, A. Porzelt, S. Inoue, *NHCs in Main Group Chemistry* Chem. Rev. 2018, 118, 9678–9842.